

```
/dev/random
```

This produces a variable length stream of random numbers.

2.3 Optimising the boot process

If you want to optimise your boot process, your first step should be to analyse how the time is being spent while your computer boots and then optimise the slow bits. You can visualise your boot process by using a tool called **bootchart**. Install it using synaptic or **apt-get**:

```
$ sudo apt-get install bootchart
```

Now reboot your machine and you will find your bootchart in **/var/log/bootchart** as a png file. Remember that bootchart will run in every boot henceforth, so don't forget to disable or uninstall it when you no longer need it. You can uninstall it using **apt-get**:

```
$ sudo apt-get remove bootchart
```

If you don't want to uninstall it, remove its SysV script from executing after startup:

```
$ cd /etc/init.d
```

```
$ sudo update-rc.d -f stop-bootchart remove
```

If you want to re-enable bootchart you may either reinstall it through the repositories or add it back to runlevels 2, 3, 4, 5:

```
$ cd /etc/init.d
```

```
$ sudo update-rc.d stop-bootchart start 99 2 3 4 5 .
```

Now that you have your boot process charted out it is time to start optimising your boot process.

Firstly you can reduce the time grub waits before it loads the kernel, the default is 5 seconds.

Now examine your bootchart and stop the programs which take too long to boot which you don't need from loading during boot. You can remove them by going to **System > Preferences > Startup Applications** and unchecking the things you don't need.

ureadahead is a process that pulls files into memory that it knows it is going to need during the boot process to make it faster. If you notice that ureadahead spends a lot of time on your bootchart it is possible that the ureadahead pack files are wrong or corrupt and that is causing it to do odd things. If this is the case you need to regenerate the ureadahead pack files, you can do so like this:

```
$ sudo rm /var/lib/ureadahead/*pack
```

The next time you reboot ureadahead will regenerate those files and you can reboot again to see if that helped.

You can also profile your boot to achieve similar results:

Profile your boot::